In the New Systems, Refilled Systems, and converted systems

Every year: add to it hours: ever year or every 2000 hours, whichever comes first.

When referring to the Service Life in table 61 , use the interval that occurs first. These coolant change intervals are only achievable with annual S·O·S Services Level 2 coolant sampling analysis.

Refer to the engine Operation and Maintenance Manual for the correct interval for replacement of the cooling system water temperature regulator.

Note: For engines that require a maximum of 20% glycol, make sure that the amount of additive in the final mix is appropriate. Example of mixing a 20% glycol solution is given in Table 63 .

Table 63

Example of Mixing Up a 20% Glycol Coolant (1)			
Total Volume of the Cooling System	Add the Following:		
	ELC Concentrate	ELI Concentrate	Water
10 Gallons	2 Gallons	0.5 Gallons	7.5 Gallons

(1) Volumes can also be in liters as long as all the volume units are consistent

Extended life coolants require the one time maintenance addition of an extender at coolant service mid-life. For commercial coolants, do NOT use an extender with a coolant unless the extender has been approved by the coolant manufacturer for use with the coolant. The coolant manufacturer is responsible to ensure compatibility and acceptable performance. Failure to follow these recommendations can result in shortened cooling system component life.

Conventional coolants require the maintenance addition of SCA throughout the expected life. For commercial coolants, do NOT use an SCA unless approved by the coolant supplier for use with the coolant. The coolant manufacturer is responsible to ensure compatibility and acceptable performance.

“ASTM D6210” require coolants that are properly dosed with SCA and that are in a properly maintained cooling system in normal service to be suitable for use for a maximum of 2 years. **The suitability for use requirement is the direct responsibility of the coolant manufacturer and SCA manufacturer.** Consult with the coolant and/or SCA manufacturer concerning the suitability of the products for use in a given application.

Cat DEAC is fully formulated and does not require a treatment with an SCA at the initial fill.

A commercial heavy-duty coolant/antifreeze that meets the “ASTM D6210”specification does not require a treatment with an SCA at the initial fill.

Commercial Heavy Duty coolants that meet only "ASTM D4985" specification are not recommended for Caterpillar applications. These coolants may not provide the performance characteristics needed for heavy duty applications. Cat coolants and the coolants detailed in Table 61 and in this Chapter are formulated to offer the required performance in Cat engines. Refer to Table 61 for Cat coolants recommendations.

Cat ELC, Cat ELI, Cat DEAC, Cat Extender, and Cat SCA are available in several container sizes. The availability of part numbers will vary by the region. Consult your Cat dealer.

In stationary and marine engine applications that do not require protection from boiling or freezing, except as noted in Table 62, Cat ELI in water or SCA and water are acceptable. **Caterpillar recommends a minimum of 7.5 percent concentration of Cat ELI in those cooling systems using Cat ELI. Caterpillar recommends a minimum of 6 percent to a maximum of 8 percent concentration of Cat SCA in those cooling systems using Cat SCA.** Distilled water or deionized water is preferred in those systems. If distilled or deionized water is not available, use water that meets or exceeds the minimum acceptable water requirements that are listed in this Special Publication, “General Coolant Information” article.

After the addition of water and proper mixing, the concentration of Cat ELI can be determined using the 360-0744 digital Brix refractometer. Refer to the tool operating manual for that refractometer for more information. The concentration of a sample of in-use Cat ELI taken from the cooling system can also be determined using this refractometer

NOTICE
All Cat engines that are equipped with a Cat NOx Reduction System require a minimum of 50 percent glycol to help prevent cavitation damage and boiling of the engine coolant. These engines include Tier 4 engines.

NOTICE
Do not use a commercial coolant/antifreeze that only meets the ASTM “D3306” or equivalent specification. This type of coolant/antifreeze is made for light duty automotive applications.

Use only the coolant/antifreeze that is recommended.

Caterpillar recommends a 50 volume percent glycol and water of proper quality. This mixture will provide optimum heavy-duty performance as a coolant/antifreeze.

PROOF

Maintain a concentration level of nitrites in the cooling system that is between 1200 ppm (70grains/US gal) and 2400 ppm (140 grains/US gal). S·O·S coolant analysis is the preferred method to check SCA concentration. Alternatively, nitrite levels can be tested with the following tools:

- 4C-9301 nitrite level test strips, English instructions only, use with glycol-based coolants such as Cat DEAC
- 286-2578 nitrite level test strips, English, French, and Spanish instructions, use with glycol-based coolants such as Cat DEAC

Frequently check the concentration of glycol in glycol-based coolant. Use a coolant/battery tester. Two products are available from Cat dealers. 245-5829 is an analogue refractometer that shows the freeze protection level of the coolant in both degrees Celsius and degrees Fahrenheit. 360-0774 is a digital Brix refractometer.

Maintain the proper concentrations of glycol and additives in the coolant. Lowering the concentration of glycol or additives will lower the ability of the coolant to protect the system from pitting, from cavitation, from erosion, and from deposits.

Do not top off the cooling system with water unless there is a specific need to adjust the water/glycol ratio. Compatible 50/50 (water/glycol) coolant is typically used and recommended when cooling system top off is required.

Clean the cooling system for the following reasons:

- Contamination of the cooling system
- Overheating of the engine
- Foaming of the coolant
- Changing from conventional heavy-duty coolant/antifreeze to Cat ELC or ELC-1 coolant

Note: Air pockets can form in the cooling system if the cooling system is filled at a rate that is greater than 20 L (5 US gal) per minute. The maximum recommended cooling system fill rate for some smaller engine models will be less. Refer to the engine Operation and Maintenance Manual for exceptions.

After you drain and refill the cooling system, operate the engine. Operate the engine without the filler cap until the coolant level stabilizes. Ensure that the coolant is maintained to the proper level.

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Extended Life Coolant

SMCS Code: 1350; 1352; 1395

Cat ELC

Caterpillar provides Cat ELC (Extended Life Coolant) for use in the following applications:

- Heavy-duty diesel engines
- Automotive applications

When Cat ELC is compared to conventional coolants, the Cat ELC anti-corrosion package is based on a different additive system. Cat ELC has been formulated with the correct amounts of additives. Superior corrosion protection is provided for all metals that are in engine cooling systems.

Cat ELC extends the service life of the coolant to 12000 service hours or 6 years. Cat ELC does not require the frequent addition of the Cat ELC Extender additive. An Extender is the only additional maintenance that is needed at 6000 service hours or one half of the Cat ELC service life.

Cat ELC is available in a 1:1 premixed cooling solution with distilled water. The premixed Cat ELC provides freeze protection to −37 °C (−34 °F). The premixed Cat ELC is recommended for the initial fill of the cooling system. The premixed Cat ELC is also recommended for topping off the cooling system.

Cat ELC Concentrate is also available. Cat ELC Concentrate can be used to lower the freezing point to −52 °C (−62 °F) for arctic conditions.

Containers of several sizes are available. Refer to this Special Publication, “Coolant Recommendations” article for available quantities and part numbers.

Note: Cat ELC can be used in most diesel and gasoline OEM engines. Cat ELC meets the performance requirements of “ASTM D6210” for heavy-duty low silicate antifreeze/coolants. Cat ELC also meets the performance requirements of “ASTM D3306” for automotive applications.

Commercial Extended Life Coolant

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids.

Inorder to be used in Cat diesel engine cooling systems at the published service intervals, select a commercial extended life coolant that meets all the requirements given in Table 64 in this Special Publication.

PROOF

Table 64

Technical Requirements for Commercial Extended Life Coolants	
Specifications	Cat EC-1 and “ASTM D6210”
Additional Requirements	Organic Additive Technology (OAT) based on a combination of a monocarboxylic acid and a dicarboxylic acid
	Phosphate, borate, and silicate free
	Minimum typical tolyltriazole level of 900 ppm for new coolants
	Minimum typical nitrite level of 500 ppm in new coolants"
Maintenance	One time addition of an extender at the mid-life of the coolant in order to maintain the coolant nitrite level between 300 - 600 ppm

Note: The Cat EC-1 specification describes the minimum requirements for extended life coolants.

Use an extender that is recommended by the EC-1 coolant supplier at mid-life of the coolant.

Commercial coolants that do not meet the minimum requirements defined in this Special Publication are not allowed for use in Cat engines.

Commercial extended life coolant used in Cat engines must meet all requirements given in Table 64 . If the ELC does meet the requirements, the service interval listed in this Special Publication may not be used. Follow the maintenance guidelines for the coolant from the supplier of the commercial extended life coolant. Follow the Cat guidelines for the quality of water and the specified coolant change interval.

Note: Coolants must be tested against the Cat EC-1 specification requirements. Coolants that only claim to meet the performance requirements of the Cat EC-1 specification may not meet all the minimum requirements.

In order to be marketed as a product that meets Cat EC-1, all Cat EC-1 specification requirements must be met. Requirements include, but are not limited to the following:

- Physical and Chemical Properties
- Compatibility Characteristics
- Bench Testing
- Field Testing

The field test includes the use of the following requirements:

- Radiator types
- Minimum field test duration
- Minimum number of diesel engines
- Cat diesel engine models of the required minimum power rating

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Extended Life Coolant Cooling System Maintenance

SMCS Code: 1350; 1352; 1395

Proper Additions to the Cat ELC (Extended Life Coolant)

NOTICE
Use only Cat products or commercial products that have passed Cat EC-1 specification for pre-mixed or concentrated coolants.

Use only Cat ELC Extender with Cat ELC.

Do NOT use conventional SCA with Cat ELC. Mixing Cat ELC with conventional coolants and/or conventional SCA reduces the Cat ELC service life.

Do NOT mix brands or types of coolant. Do NOT mix brands or types of SCA and/or brands or types of extenders. Different brands or types may use different additive packages to meet the cooling system requirements. Different brands or types may not be compatible.

Failure to follow the recommendations can reduce cooling system component life unless appropriate corrective action is performed.

In order to maintain the correct balance between the antifreeze and the additives, maintain the recommended concentration of Cat ELC. Lowering the proportion of antifreeze lowers the proportion of additive. The coolant will not be able to protect the system from pitting, cavitation, erosion, and deposits.

PROOF

During daily maintenance, use the premixed Cat ELC as a cooling system top-off. Check the specific gravity of the coolant system with the 245-5829 Coolant/Battery Tester/Refractometer. This tester gives readings that are immediate and accurate in both degrees Celsius and degrees Fahrenheit. Use Cat ELC Concentrate to restore the proper glycol concentration in the coolant system. Add the concentrate before the engine is exposed to freezing temperatures.

NOTICE
Do not use a conventional coolant to top-off a cooling system that is filled with Cat ELC.

Do not use conventional SCA. Only use Cat ELC Extender in cooling systems that are filled with Cat ELC.

Cat ELC Extender

Cat ELC Extender is added to the cooling system halfway through the Cat ELC service life. Treat the cooling system with Cat ELC Extender at 6000 hours or one half of the coolant service life. Refer to your machine Operation and Maintenance Manual for exceptions. Refer to the “Part Number of Coolant” table in this Special Publication, “Coolant Recommendations” article for available quantities and part numbers.

Use the formula in Table 66 to determine the proper amount of Cat ELC Extender for your cooling system. Refer to Operation and Maintenance Manual, “Refill Capacities and Recommendations” in order to determine the capacity of the cooling system.

Table 65

Formula For Adding Cat ELC Extender To Cat ELC
V × 0.02 = X
V is the total capacity of the cooling system.
X is the amount of Cat ELC Extender that is required.

Table 66 is an example for using the formula that is in Table 65 . This example is based on the typical capacity of a D8R Track-Type Tractor.

Table 66

Example Of The Equation For Adding Cat ELC Extender To Cat ELC		
Total Volume of the Cooling System (V)	**Multiplication Factor**	**Amount of Cat ELC Extender that is Required (X)**
94.6 L (25.0 US gal)	× 0.02	1.9 L (0.5 US gal)

NOTICE
When using Cat ELC, do not use conventional SCAs, or, if equipped, SCA maintenance elements. In order to avoid SCA contamination of an ELC system, remove the SCA element base and plug off or bypass the coolant lines.

Cat ELC Cooling System Cleaning

Note: If the cooling system is already using Cat ELC, cleaning agents are not required at the specified coolant change interval. Cleaning agents are only required if the system has been contaminated by the addition of some other type of coolant or by cooling system damage.

Clean water is the only cleaning agent that is required when Cat ELC is drained from a properly maintained cooling system.

After the cooling system is drained and after the cooling system is refilled, operate the engine while the cooling system filler cap is removed. Operate the engine until the coolant level reaches the normal operating temperature and until the coolant level stabilizes. As needed, add the coolant mixture in order to fill the system to the proper level.

Recycling Cat ELC

Cat ELC can be recycled into conventional coolants. The drained coolant mixture can be distilled in order to remove the ethylene glycol and the water. The ethylene glycol and the water can be reused. The distilled material does not contain the additives that are classified as either Cat ELC or Cat DEAC. Consult your Cat dealer for more information. Recycled coolants should meet the most current revision level of “ASTM D6210”.

Changing to Cat ELC

To change from heavy-duty coolant/antifreeze to the Cat ELC, perform the following steps:

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, “Caterpillar Dealer Service Tool Catalog” and to Special Publication, PECJ0003, “Cat Shop Supplies and Tools” for tools and supplies suitable to collect and contain fluids on Cat products.

Dispose of all fluids according to applicable regulations and mandates.

PROOF

1. Drain the coolant into a suitable container.

2. Dispose of the coolant according to local regulations.

3. If equipped, remove the empty SCA maintenance element and remove the element base. Plug the coolant lines or bypass the coolant lines.

NOTICE
Do not leave an empty SCA maintenance element on a system that is filled with Cat ELC.

The element housing may corrode and leak causing an engine failure.

Remove the SCA element base and plug off or by-pass the coolant lines.

4. Flush the system with clean water in order to remove any debris.

5. Use Cat Quick Flush Cooling System Cleaner for cooling systems in order to clean the system. Cat Quick Flush Cooling System Cleaner is available in various sizes. Part numbers are 4C-4609 (0.5 L (0.125 US gal)) through 4C-4613 (208.2 L (55 US gal)). Follow the instructions on the label using a 6-10% concentration of cleaner in water.

6. Drain the cleaner into a suitable container. Flush the cooling system with clean water.

Note: Deposits that remain in the system may be loosened and removed by the Cat ELC.

7. In systems with heavy deposits, disconnect the hoses. Clean the deposits and debris from the hoses and the fittings. Install the hoses and tighten the hose fittings. Refer to Specifications, SENR3130, “Torque Specifications” for the proper torques. Pipe threads may also need to be cleaned and sealed. Seal the threads with 5P-3413 Pipe Sealant.

Note: Replace hoses that are cracked, soft, or that have other signs of damage. Tighten all hose clamps and fittings.

8. Fill the cooling system with clean water and operate the engine until the engine is warmed to 49 °C to 66 °C (120 °F to 151 °F).

NOTICE
Improper or incomplete rinsing of the cooling system can result in damage to copper and other metal components.

To avoid damage to the cooling system, make sure to completely flush the cooling system with clear water. Continue to flush the system until all signs of the cleaning agent are gone.

9. Drain the cooling system into a suitable container and flush the cooling system with clean water.

NOTICE
The cooling system cleaner must be thoroughly flushed from the cooling system. Cooling system cleaner that is left in the system will contaminate the coolant. The cleaner may also corrode the cooling system.

10. Repeat Steps 8 and 9 until the system is clean.

11. Fill the cooling system with the Cat ELC.

12. Operate the engine until the engine is warmed. While the engine is running, inspect the engine for leaks. Tighten hose clamps and connections in order to stop any leaks.

13. Attach the Special Publication, PMEP5027, “Label” to the cooling system filler for the engine in order to indicate the use of Cat ELC.

Note: Clean water is the only flushing agent that is required when Cat ELC is drained from a properly maintained cooling system.

Cat ELC Cooling System Contamination

NOTICE
Mixing Cat ELC with other products reduces the effectiveness of the Cat ELC and shortens the Cat ELC service life. Use only Cat products or commercial products that have passed the Cat EC-1 specification for premixed or concentrate coolants. Use only Cat ELC Extender with Cat ELC. Do NOT mix brands or types of coolants. Failure to follow these recommendations can result in shortened cooling system component life.

Cat ELC cooling systems can withstand contamination to a maximum of 10 percent of conventional heavy-duty coolant/antifreeze and/or SCA. The advantages of Cat ELC are reduced above 10 percent. If the contamination exceeds 10 percent of the total system capacity, perform ONE of the following procedures:

PROOF

- If the cooling system contamination is caused by cooling system damage, follow the procedures under the “Changing to Cat ELC” heading. Also, follow the procedures under the “Changing to Cat ELC” heading if the engine has been operated since being contaminated with more than 10 percent conventional heavy-duty coolant/ antifreeze and/or SCA. Certain types of cooling system contamination may require disassembly of the cooling system and manual cleaning of system components.
- If the cooling system is contaminated with more than 10 percent conventional heavy-duty coolant/ antifreeze and/or SCA, but the engine has not been operated, drain the cooling system into a suitable container. Dispose of the coolant according to local regulations. Thoroughly flush the system with clean water. Fill the system with the Cat ELC.
- Maintain the system as a conventional DEAC (Diesel Engine Antifreeze/Coolant) or other conventional coolant. Submit a coolant sample for Level 2 S·O·S Coolant Analysis to determine the coolant condition. Maintain a 3 to 6 percent SCA concentration in the coolant. Change the coolant at the interval that is recommended for Cat DEAC or at the interval that is recommended for the conventional commercial coolants.

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Extended Life Inhibitor (ELI)

SMCS Code: 1350; 1352; 1395

Cat Extended Life Inhibitor (ELI) is water-based coolant that does not contain glycol. Cat ELI is for applications that do not require freeze protection. Exceptions are listed here. Failure to follow these recommendations can or will result in failures.

Cat ELI is a corrosion inhibitor concentrate that is mixed to approximately 7.5 % by volume with water. Cat ELI has the following characteristics:

- Based on the same organic additive technology that is used in Cat Extended Life Coolant (ELC)
- Does not contain glycol. Designed for use in applications that do not require freeze protection.
- Provides superior corrosion and cavitation protection compared to SCA mixed with water.
- Provides an extended drain interval of up to 6 years or 12,000 hours. The drain interval may be longer as determined by using Cat S·O·S coolant analysis.
- Requires little maintenance compared to conventional SCA mixed with water.

Cat ELI can replace SCA/Water coolant in engine applications that do not require freeze protection.

Exceptions for use of water-based ELI or SCA coolants are listed in this Special Publication, “Coolant Recommendations - Special Requirements Table”.

Additional information is available from your Cat Dealer. Refer to Information Release Memo, PELJ1212, “Cat ELI – PERFORMANCE LIKE Cat ELC FOR APPLICATIONS NOT REQUIRING FREEZE PROTECTION”. Refer to Table 67 for information about Cat ELI.

PROOF

Table 67

Part Number	Container Size	Volume of Finished Coolant Produced
351-9431	3.8 L (1 US gal)	50.5 L (13.3 US gal)
351-9432	20 L (5.3 US gal)	267 L (70.5 US gal)
351-9433	208 L (55 US gal)	2773 L (733 US gal)
366-2753 (1)	1000 L (264 US gal)	13333 L (3523 US gal)

(1) NACD and LACD only

Mixing Cat ELI

The recommended water for mixing with Cat ELI concentrate is distilled or deionized water. Water must meet requirements of ASTM 1193, “Type IV Reagent Water Specification”. If distilled or deionized water is not available, water should meet the “Caterpillar Minimum Acceptable Water Requirements” provided in this Special Publication.

To ensure a proper concentration, the preferred method is to mix Cat ELI concentrate with water. Then, add the mixed coolant to the engine cooling system. Add the proper amounts of water and Cat ELI into a clean container and mix thoroughly by manual stirring or mechanical agitation.

If the preferred method cannot be performed, a Cat ELI mixture can be made by adding Cat ELI concentrate directly into engine cooling system. Add good quality water until the dilution level is approximately 7.5%. Adequate mixing is attained by operating the engine for at least 30 minutes.

Appropriate mixing rates for available ELI container sizes are provided in Table 67 .

After the addition of water and proper mixing, the concentration of Cat ELI can be determined using the 360-0774 Digital Brix Refractometer.

Changing to Cat ELI

For cooling systems previously running Cat ELC or an extended life coolant that meets Cat EC-1 requirements, drain the cooling system and flush with water. Then refill the cooling system with a mixture of 7.5% Cat ELI in water that meets the “Caterpillar Minimum Acceptable Water Requirements”.

For cooling systems previously running a conventional heavy-duty coolant or a water/SCA mixture, follow the steps listed in this Special Publication, “Changing to Cat ELC”. Then refill the cooling system with a mixture of 7.5% Cat ELI in water that meets the “Caterpillar Minimum Acceptable Water Requirements”.

Cat ELI Maintenance

Maintenance of Cat ELI is similar to Cat ELC. A coolant sample should be submitted for S·O·S Level 2 Coolant Analysis after the first 500 hours of operation and then annually thereafter.

Cat ELC Extender should be added at the midpoint of service life (typically 6,000 hours), or as recommended by S·O·S Coolant Analysis results.

Analysis and interpretation of Cat ELI S·O·S results is similar to the analysis and interpretation of Cat ELC. There will be no glycol and glycol oxidation products, which do not apply to Cat ELI.

The concentration of a sample of in-use Cat ELI taken from the cooling system can also be determined using the 360-0774 Digital Brix Refractometer.

PROOF

Note: Clean water is the only flushing agent that is required when Cat ELI is drained from a properly maintained cooling system.

Mixing Cat ELI and Cat ELC

Since Cat ELI and Cat ELC are based on the same corrosion inhibitor technology, Cat ELI can be mixed with Cat ELC. Mixing may be desired when only low level of freeze protection is required. Consult your local Cat dealer to ensure proper mixing of the products to provide adequate freeze protection and corrosion protection.

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Diesel Engine Antifreeze/ Coolant
(DEAC)

SMCS Code: 1350; 1352; 1395

Follow the maintenance information provided in this Special Publication, “Coolant and General Maintenance Recommendations”section.

Cat recommends using Cat DEAC (Diesel Engine Antifreeze/Coolant) for cooling systems that require a high performance conventional heavy-duty coolant/ antifreeze. Cat DEAC is alkaline single-phase ethylene glycol type antifreeze that contains corrosion inhibitors and antifoam agents.

Cat DEAC is formulated with the correct amount of Cat SCA (Supplemental Coolant Additive). Do not use Cat SCA at the initial fill when Cat DEAC is used at the recommended 50 percent by volume glycol or higher concentration with recommended water.

Containers of several sizes are available. Consult your Cat dealer for the part numbers.

If concentrated Cat DEAC is used, Caterpillar recommends mixing the concentrate with distilled water or with deionized water. If distilled water or deionized water is not available, use water which has the required properties. For the water properties, see this publication, “General Coolant Information”.

Note: Mix the concentrated Cat DEAC and water thoroughly prior to filling the cooling system.

SCA is added to cooling systems that use conventional coolants for one of three reasons:

- to precharge a new conventional coolant that is not fully Formulated. Cat DEAC is fully formulated and does not require SCA
- to provide corrosion protection in water/SCA cooling systems
- to recharge an in service conventional coolant on a maintenance basis to maintain proper inhibitor levels

Conventional coolants typically require SCA additions at approximately 250-300 service hours. The use of SCA helps to prevent the following conditions from occurring:

- Corrosion
- Formation of mineral deposits
- Cavitation erosion of the cylinder liners
- Foaming of the coolant

Containers of Cat SCA are available in several sizes. Refer to this Special Publication, “Conventional Coolant/Antifreeze Cooling System Maintenance” article for available quantities and part numbers.

Note: Do not exceed a 6 percent maximum concentration of SCA. Maintain a 3 to 6 percent SCA concentration in the coolant.

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Supplemental Coolant Additive

SMCS Code: 1350; 1352; 1395

Supplemental Coolant Additive (SCA) – SCA is a general term for an inhibitor package that is added to a cooling system. SCA is added for one of three reasons:

- to precharge a new coolant that is not fully formulated
- to provide corrosion protection in water/SCA cooling systems
- to recharge an in service conventional coolant on a maintenance basis to maintain proper inhibitor levels

Conventional coolants typically require SCA additions at approximately 250-300 service hours.

The use of SCA (supplemental coolant additive) helps to prevent the following conditions from occurring:

- Corrosion
- Formation of mineral deposits
- Cavitation erosion of the cylinder liners
- Foaming of the coolant

PROOF

Cat DEAC (Diesel Engine Antifreeze/Coolant) is formulated with the correct level of Cat SCA. When the cooling system is initially filled with Cat DEAC, adding more Cat SCA is not necessary. To ensure that the correct amount of Cat SCA is in the cooling system, test the concentration on a scheduled basis. Testing the concentration of nitrites and silicates through the S·O·S analysis is the best indicator of the amount of SCA in the coolant.

Maintain a concentration level of nitrites in the cooling system that is between 1200 ppm (70grains/US gal) and 2400 ppm (140 grains/US gal). S·O·S coolant analysis is the preferred method to check SCA concentration. Alternatively, nitrite levels can be tested with the following tools:

- 4C-9301 nitrite level test strips, English instructions only, use with glycol-based coolants such as Cat DEAC
- 286-2578 nitrite level test strips, English, French, and Spanish instructions, use with glycol-based coolants such as Cat DEAC

For the maintenance schedule, refer to the Operation and Maintenance Manual, “Maintenance Interval Schedule” (Maintenance Section) of the specific engine.

Containers of Cat SCA are available in several sizes. Refer to this Special Publication, “Conventional Coolant/Antifreeze Cooling System Maintenance” article for available quantities and part numbers.

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Conventional Coolant/ Antifreeze Cooling System Maintenance

SMCS Code: 1350; 1352; 1395

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids.

NOTICE
Use Only Approved SCAs. Conventional coolants require the maintenance addition of SCA throughout their expected life. Do NOT use an SCA with a coolant unless specifically approved by the coolant supplier for use with their coolant. It is the responsibility of the coolant manufacturer to ensure compatibility and acceptable performance.

Failure to follow the recommendations can result in shortened cooling system component life.

Cat SCA is compatible with Cat DEAC. If you use non-Cat brand conventional coolants, consult with the coolant manufacturer for information on a compatible SCA.

NOTICE
Do NOT mix brands or types of coolant. Do NOT mix brands or types of SCA and/or brands or types of extenders. Different brands or types may use different additive packages to meet the cooling system requirements. Different brands or types may not be compatible.

Failure to follow the recommendations can reduce cooling system component life unless appropriate corrective action is performed.

If non-Cat conventional coolants that meet Cat minimum published requirements are used, confirm with the coolant manufacturer before using Cat SCA that Cat SCA is acceptable for use with the coolant. Do not exceed the 6 percent maximum concentration. Check the concentration of SCA with a SCA test kit, or check the concentration of SCA with Cat S·O·S coolant analysis.

Cat engine cooling systems should be tested at 250 hour intervals or at the PM Level 1 intervals for the concentration of SCA (Supplemental Coolant Additive).

Note: Acceptable conventional heavy-duty coolant/antifreeze of all types REQUIRE periodic additions of SCA.

Follow the maintenance information provided in this Special Publication, “Coolant and General Maintenance Recommendations” section.

Note: Specific engine applications may require maintenance practices to be periodically evaluated to maintain the engine cooling system properly.

The availability of part numbers will vary from one region to another region.

Do not exceed the 6 percent maximum concentration. Check the concentration of SCA with a SCA test kit, or check the concentration of SCA with Cat S·O·S coolant analysis.

Adding the SCA to Pure Water or a Pure Water/Glycol Mixture at Initial Fill

For initial fill, Cat SCA added to pure water provides corrosion protection of the system. Ensure that the water meets quality requirements detailed in the Table “Cat Minimum Acceptable Water Requirements”, found in this Special Publication, “General Coolant Information”. Also, ensure that the water or water/glycol mixture have not had additives previously added. Use the equation in Table 68 to determine the volume of the Cat SCA that is needed. The initial concentration of the SCA is recommended to be 7 percent. Once the coolant has been in service, maintain the SCA to a 3 to 6 percent range. Refer to the maintenance section for details.

PROOF

Table 68

Equation For Adding The Cat SCA To Water At The Initial Fill
V × 0.07 = X
V is the total volume of the cooling system.
X is the amount of Cat SCA that is required.

Table 69 is an example for using the equation that is in Table 68 .

Table 69

Example Of The Equation For Adding The Cat SCA To Water At The Initial Fill		
Total Volume of the Cooling System (V)	**Multiplication Factor**	**Amount of Cat SCA that is Required (X)**
946 L (250 US gal)	× 0.07	66 L (18 US gal)

Adding the SCA to Conventional Coolant/Antifreeze For Maintenance

Conventional heavy-duty coolant/antifreeze of all types REQUIRE periodic additions of a supplemental coolant additive.

Test the coolant/antifreeze periodically for the concentration of SCA. For the interval, see the Operation and Maintenance Manual, “Maintenance Interval Schedule”, for your engine. SCA test kits are available from your Cat dealer. Test the concentration of SCA or submit a coolant sample to your Cat dealer. Refer to this Special Publication, “S·O·S Services Coolant Analysis”, article.

Additions of SCA are based on the results of the test or based on the results of the coolant analysis. The size of the cooling system determines the amount of SCA that is needed.

Table 70

Equation for the Addition of Cat SCA To Water For Maintenance
V × 0.023 = X
V is the total volume of the cooling system.
X is the amount of Cat SCA that is required.

Table 71 is an example for using the equation that is in Table 70 .

Table 71

Example of the Equation for the Addition of Cat SCA To Water For Maintenance		
Total Volume of the Cooling System (V)	**Multiplication Factor**	**Amount of Cat SCA that is Required (X)**
946 L (250 US gal)	× 0.023	22 L (6 US gal)

Note: Specific engine applications may require maintenance practices to be periodically evaluated to maintain the engine cooling system properly.

Cleaning the System of Heavy-Duty Coolant/Antifreeze

Before Cat SCA can be effective, the cooling system must be free from rust, scale, and other deposits. Preventive cleaning helps avoid downtime caused by expensive out-of-service cleaning required for extremelydirty and neglected cooling systems.

Cat Cooling System Cleaners

- Dissolves or depresses mineral scales, corrosion products, light oil contaminations, and sludges
- Cleans the cooling system after used coolant is drained or before the cooling system is filled with new coolant
- Cleans the cooling system whenever the coolant is contaminated or whenever the coolant is foaming
- The “Standard” version of the Cat Cooling System Cleaners clean the cooling system while still in service.
- Reduces downtime and cleaning costs
- Helps avoid costly repairs from pitting and other internal problems caused by improper cooling system maintenance
- Can be used with glycol-based antifreeze
- For the recommended service interval, refer to the Operation and Maintenance Manual, “Maintenance Interval Schedule” for your engine.

Cat Standard Cooling System Cleaners are designed to clean the cooling system of harmful scale and corrosion without removing the engine from service. The cleaners, both “Standard” and “Quick Flush” can be used in all Cat engine cooling systems. Consult your Cat dealer for part numbers.

PROOF

Note: These cleaners must not be used in systems that have been neglected or that have heavy scale buildup. These systems require a stronger commercial solvent that is available from local distributors.

Prior to performing a cleaning of the cooling system, take a 1-liter (1-quart) sample of coolant from the engine while in operation into a clear container. Take the sample shortly after start-up while the coolant is not yet hot. The coolant should be adequately mixed by the water pump. Allow the sample to sit for 2 hours. If a visible oil layer is present, neither the Standard cleaners nor Quick Flush cleaner will be fully effective. First, drain the coolant and then perform the procedure given below (using non-foaming dish detergent), followed by the procedure for the Quick Flush cleaner.

Procedure for Cleaning an Oil Contaminated Cooling System

1. Drain the cooling system.

2. Fill the cooling system with acceptable quality water.

Note: Refer to the “Caterpillar Minimum Acceptable Water Quality Requirements” in this Special Publication.

3. Start the engine and run the engine until the thermostat opens.

4. Add a non-foaming detergent containing tripolypentasodium phosphate (TSP) to equal approximately 2-3% cooling system capacity.

Note: Pre-dissolve the detergent in approximately 19 L (5 US gallons) of acceptable quality water. Add this mixture directly to the cooling system and top off the cooling system with water.

5. Run the engine for at least 30 minutes. Drain the cooling system.

6. Remove a small sample of the detergent solution from the cooling system. Allow the solution to sit for at least 30 minutes and check for signs of a visible oil layer on top. If oil is still present, repeat the procedure.

Note: Corrosion of the metal can occur if the detergent solution is left in the cooling system for longer than 1 hour.

7. Flush the cooling system, if there is no visible oil layer in the solution. Fill the cooling system with acceptable quality water. Run the engine for 20 minutes and then drain the water.

8. Perform the cleaning procedure found in this Special Publication for “Cat Cooling System Cleaner - Quick Flush”, if additional removal of scale, rust, and inhibitor deposits from the previous coolant is needed.

9. If additional cleaning is not needed, refill the cooling system with new coolant.

Table 72

Cat Cooling System Cleaner for use with Quick Flush Method	
Cleaner part number	Size of Container
4C-4609	0.5 L (0.125 US gal)
4C-4610	1 L (0.25 US gal)
4C-4611	3.8 L (1 US gal)
4C-4612	19 L (5 US gal)
4C-4613	208 L (55 US gal)

Procedure for using Cat Cooling System Cleaner - Quick Flush

1. Drain the cooling system.

2. Refill the cooling system to 90-94% capacity with acceptable quality water.

Note: Refer to the “Caterpillar Minimum Acceptable Water Quality Requirements” as found in this Special Publication.

3. Top off the cooling system with Cat Cooling System Cleaner - Quick Flush cleaner so that the solution is 6-10% cleaner.

4. Run the engine for 1.5 hours.

5. After allowing adequate time for the system to cool, drain the cooling system.

Note: Corrosion of the metals can occur if the Quick Flush cleaning solution is in the cooling system for longer than 2 hours.

6. Flush the cooling system.

a. Fill the cooling system with acceptable quality water.

b. Run the engine for 20 minutes.

c. Drain the water.

7. Refill the cooling system with new coolant.

PROOF

Table 73

Cat Cooling System Cleaner - Standard		
Part Number	Volume	Availability
6V-4511	1.89 L (0.5 US gal)	North and South America

Procedure for using Cat Cooling System Cleaner - Standard

1. If necessary, drain off coolant from the cooling system for the amount of the cleaner.

2. Add 1 bottle (1.89 L (0.5 US gal)) of Cat Cooling System Cleaner - Standard for each 30 L of cooling system capacity. Add the cleaner directly to the coolant.

3. Run the engine as usual.

4. After 30 days, drain the cooling system.

5. Flush the cooling system.

a. Fill the cooling system with acceptable quality water.

b. Run the engine for 20 minutes.

c. Drain the water.

6. Refill the cooling system with new coolant.

Recycling Cat DEAC

Cat DEAC can be recycled. The drained coolant mixture can be distilled to separate the ethylene glycol and water. The ethylene glycol and the water can be reused. The distilled material does not contain the additives that are classified as either Cat ELC or Cat DEAC. Consult your Cat dealer for more information.

When recycled coolants are used, use only coolants that have been recycled from extended life, heavy-duty, or automotive coolants. Use coolants that were originally manufactured from virgin ethylene or propylene glycol.

Recycled coolants should meet the latest revision of “ASTM D6210”.

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Commercial Heavy-Duty Coolant/Antifreeze and SCA (Supplemental Coolant Additive)

SMCS Code: 1350; 1352; 1395

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids.

NOTICE
Do NOT mix brands or types of SCA. Do NOT mix SCAs and extenders.

Failure to follow the recommendations can result in shortened cooling system component life.

NOTICE
Use Only Approved SCAs. Conventional coolants require the maintenance addition of SCA throughout their expected life. Do NOT use an SCA with a coolant unless specifically approved by the coolant supplier for use with their coolant. It is the responsibility of the coolant manufacturer to ensure compatibility and acceptable performance.

Failure to follow the recommendations can result in shortened cooling system component life.

Follow the maintenance information provided in the “Coolant and General Maintenance Recommendations” section in this Special Publication.

Select a commercial diesel engine antifreeze coolant that meets all the requirements given in Table 74 . The table contains the requirements for coolant to meet the published service intervals.

The provided requirements are applicable to finished coolants and not for the concentrates. When concentrated coolant/antifreeze is mixed, Cat recommends mixing the concentrate with distilled water or with deionized water. If distilled water or deionized water is not available, water which has the required properties may be used. For the water properties, refer to this Special Publication, “General Coolant Information” article.

PROOF

Coolant/antifreezes for heavy-duty applications that meet “ASTM D6210” do not require treatment with SCA at the initial fill. Use the recommended 1:1 or higher concentration with recommended water. Treatment with SCA is required on a maintenance basis.

The SCA manufacturer is responsible for ensuring the SCA is compatible with water meeting the “Caterpillar Minimum Acceptable Water Quality Requirements” as found in this Special Publication, and “ASTM D6210-08, Table X1.1”. The coolant manufacturer and the SCA manufacturer are responsible to ensure that the products will not cause cooling system harm.

Do not mix brands or types of coolants with different brands or types of SCA or extender.

If using non Cat coolants, refer to the coolant manufacturer for information on a compatible SCA.

Treat the compatible commercial coolant with 3 to 6 percent Cat SCA by volume. Maintain a 3 to 6 percent concentration level of SCA in the cooling system. For more information, refer to this Special Publication, “Conventional Coolant/Antifreeze Cooling System Maintenance” article.

Table 74

Technical Requirements for Commercial Diesel Engine Antifreeze Coolants	
Specifications	ASTM D6210-08
Additional Requirements	Silicon: 100 ppm minimum to 275 ppm maximum Nitrites: maintained at 1200ppm (70 grains/US gal) minimum to 2400 ppm (140 grains/US gal) maximum Cat SCA at 3 to 6 percent (if Cat SCA is added)
Maintenance	ASTM D6210-08 Add compatible SCA at maintenance intervals Clean and flush the cooling system at drain intervals

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Water/SCA (Supplemental Coolant Additive)

SMCS Code: 1350; 1352; 1395

Cat SCA can be added to water of the recommended quality to form a Water/SCA finished coolant. SCA/Water finished coolant is glycol free. SCA/Water finished coolant is for engine applications that do not require freeze protection.

Exceptions are listed in this Special Publication, “Coolant Recommendations - Special Requirements”. Failure to follow these recommendations can or will result in failures.

NOTICE
Never use water alone as a coolant. Water alone is corrosive at engine operating temperatures. In addition, water alone does not provide adequate protection against boiling or freezing.

In engine cooling systems that use water alone, Caterpillar recommends the use of Cat SCA. Cat SCA helps to prevent the following conditions from occurring:

- Corrosion
- Formation of mineral deposits
- Cavitation erosion of the cylinder liner
- Foaming of the coolant

If Cat SCA is not used, select a fully formulated commercial SCA. The commercial SCA must provide a minimum of 2400 mg/L or 2400 ppm (140 grains/US gal) of nitrites in the final coolant mixture.

PROOF

The quality of the water is an important factor in this type of cooling system. Distilled water or deionized water is recommended for use in cooling systems. If distilled water or deionized water is not available, use water that meets or exceeds the minimum requirements that are listed in the table for recommended water properties in this Special Publication, “General Coolant Information” topic.

A cooling system that uses a mixture of SCA and water only needs more SCA. The SCA concentration in a cooling system that uses SCA and water should be 6 to 8 percent by volume.

Maintain the Cat SCA in the same way as you would maintain a cooling system that uses heavy-duty coolant/antifreeze. Adjust the maintenance for the amount ofCat SCA additions.

Adding the Cat SCA to Water at the Initial Fill

Use the equation that is in this Special Publication, “Conventional Coolant/Antifreeze Cooling System Maintenance” to determine the amount of Cat SCA that is required at the initial fill. This equation is for a mixture of only Cat SCA and water.

Adding the Cat SCA to Water for Maintenance

For the recommended service interval, refer to the Operation and Maintenance Manual, “Maintenance Interval Schedule” for your engine.

Submit a coolant sample to your Cat dealer. See this Special Publication, “S·O·S Services Coolant Analysis” topic.

Additions of Cat SCA are based on the results of the coolant analysis. The size of the cooling system determines the amount of Cat SCA that is required.

Use the equation that is in this Special Publication, “Conventional Coolant/Antifreeze Cooling System Maintenance” to determine the amount of Cat SCA that is required for maintenance, if necessary:

Note: Specific engine applications may require maintenance practices to be periodically evaluated in order to maintain properly the engine cooling system.

SCA and part numbers are available from your Cat dealer.

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S·O·S Services Coolant Analysis

SMCS Code: 1350; 1395; 7542

Testing the engine coolant is important to ensure that the engine is protected from internal cavitation and corrosion. The analysis also tests the ability of the coolant to protect the engine from boiling and freezing. S·O·S coolant analysis can be done at your Caterpillar dealer. Caterpillar S·O·S coolant analysis is the best way to monitor the condition of your coolant and your cooling system. S·O·S coolant analysis is a program that is based on periodic samples.

NOTICE
Do not use the same vacuum sampling pump for extracting oil samples that is used for extracting coolant samples.

A small residue of either type sample may remain in the pump and may cause a false positive analysis for the sample being taken.

Always use a separate pump for oil sampling and a separate pump for coolant sampling.

Failure to do so may cause a false analysis which could lead to customer and dealer concerns.

New Systems, Refilled Systems, and Converted Systems

Perform an S·O·S coolant analysis (Level 2) at the following maintenance intervals.

- Initial 500 service hours
- Every Year or every 2000 hours, whichever comes first

PROOF

Perform this analysis at the interval that occurs first for new systems, for refilled systems, or for converted systems that use Cat ELC (Extended Life Coolant) or use Cat DEAC (Diesel Engine Antifreeze/Coolant). This 500 hour check will also check for any residual cleaner that may have contaminated the system.

Recommended Interval for S·O·S Services Coolant Sample

The following table contains the recommended sampling interval for all coolants that meet Cat EC-1 (Engine Coolant specification - 1). This is also the recommended sampling interval for all conventional heavy-duty coolant/antifreeze.

The Level 2 Coolant Analysis should be performed if a problem is suspected or identified.

Table 75

Recommended Interval		
Type of Coolant	**Level 1**	**Level 2**
Cat DEAC and Conventional Heavy-Duty Coolants	Every 250 hours	Yearly
Cat ELC and Commercial EC-1 coolants	Optional or every 500 hours	Yearly or every 500 hours

Note: Check the SCA (Supplemental Coolant Additive) of the conventional coolant at every oil change or at every 250 hours. Perform this check at the interval that occurs first.

Refer to your machine OMM for recommendations specific to your machine.

S·O·S Services Coolant Analysis (Level 1)

A coolant analysis (Level 1) is a test of the properties of the coolant.

The following properties of the coolant are tested:

- Glycol concentration for freeze protection and boil protection
- Ability to protect from erosion and corrosion
- pH
- Conductivity
- Visual analysis
- Odor analysis

The results are reported, and appropriate recommendations are made.

S·O·S Services Coolant Analysis (Level 2)

A coolant analysis (Level 2) is a comprehensive chemical evaluation of the coolant. This analysis is also a check of the overall condition of the cooling system.

The S·O·S coolant analysis (Level 2) has the following features:

- Full coolant analysis (Level 1)
- Identification of metal corrosion and of contaminants
- Identification of buildup of the impurities that cause corrosion
- Identification of buildup of the impurities that cause scaling
- Determination of the possibility of electrolysis within the cooling system of the engine

The results are reported, and appropriate recommendations are made.

For more information on S·O·S coolant analysis, consult your Caterpillar dealer.

PROOF

Exhaust Aftertreatment Fluids Specifications

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Diesel Exhaust Fluid (DEF) (For Use with SCR-equipped engines)

SMCS Code: 108K

General Information

Diesel Exhaust Fluid (DEF) is a liquid that is injected into the exhaust system of engines equipped with Selective Catalytic Reduction (SCR) systems. SCR reduces emissions of nitrogen oxides (NOx) in diesel engine exhaust.

Diesel Exhaust Fluid (DEF) is available in many brands, including those brands that carry the AdBlue or API certification. DEF is also generically referred to as urea.

In engines equipped with SCR emissions reduction system, DEF is injected in controlled amounts into the engine exhaust stream. At the elevated exhaust temperature, urea in DEF is converted to ammonia. The ammonia chemically reacts with NOx in diesel exhaust in the presence of the SCR catalyst. The reaction converts NOx into harmless nitrogen (N2) gas and water vapor (H2O).

Note: Follow all applicable industry standards and all applicable governmental, environmental, safety guidelines, practices, regulations, and mandates.

Note: These general recommendations and guidelines concerning maintenance and care of DEF and DEF storage systems are not intended to be all inclusive. Discuss proper DEF safety, health, handling, and maintenance practices with your DEF supplier. Use of these general recommendations and guidelines does not lessen the responsibility of the engine owner and/or DEF supplier to follow all industry standard practices for DEF storage and for DEF handling.

DEF Recommendations for SCR Aftertreatment Systems

For use in Cat engines, DEF must meet all the requirements defined by the “ISO 22241-1” standard.

The caps of DEF tanks are typically blue as recommended by the “ISO 22241-4” Standard. Fill DEF only in clearly marked DEF tanks that have the blue cap.

Caterpillar recommends the use of DEF available through the Cat parts ordering system for use in Cat engines equipped with SCR systems. Refer to Table 76 for part number information:

Table 76

Cat Part Number (32.5% Urea)	Container Size
350-8733	2.5 gal bottle
350-8734	1000-L tote

In North America, commercial DEF that is API approved and meets all the requirements defined in “ISO 22241-1” may be used in Cat engines that are equipped with SCR systems.

Outside of North America, commercial DEF that meets all requirements defined in “ISO 22241-1” may be used in Cat engines that are equipped with SCR systems.

The supplier should provide documentation to prove that the DEF is compliant with the requirements of “ISO 22241-1”.

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids.

NOTICE
Do not use agriculture grade urea solutions. Do not use any fluids that do not meet “ISO 22241-1” Requirements in SCR emissions reduction systems. Use of these fluids can result in numerous problems including damage to SCR equipment and a reduction in NOx conversion efficiency.

DEF is a solution of solid urea that is dissolved in deionized water to produce a final concentration of 32.5% urea. Most SCR systems are designed to operate only on DEF concentration of 32.5 percent. DEF solution of 32.5% urea has the lowest attainable freeze point of −11.5° C (11.3° F). DEF concentrations that are higher or lower than 32.5% have higher freeze points. DEF dosing systems and “ISO 22241-1” specifications are designed for a solution that is approximately 32.5%.

Cat offers refractometers that can be used to measure DEF concentration. Refer to Table 77 for the part numbers. Follow the instructions provided with the instruments.

Table 77

Cat DEF Refractometers		
Refractometer Part Numbers	360-0774	Digital display, easy to use, and multifunctional (DEF, coolant, battery fluid, and more)

(continued)

PROOF

(Table 77, contd)

	431-7087	Analog, specific to DEF, and requires a multi-step test procedure

DEF Guidelines

DEF solution is typically colorless and clear. Changes to color or clarity are indicators of quality issues. Quality of DEF can degrade when stored and handled inappropriately or if DEF is not protected from contamination. Details are provided below.

If quality issues are suspected, testing of DEF should focus on urea percentage, alkalinity as NH3 and biuret content. DEF that does not pass all these tests or that is no longer clear should not be used.

Note: Caterpillar strongly recommends that customers purchase the pre-mixed DEF urea solution from a reputable supplier. The DEF must satisfy all the specifications of quality given in this chapter of this Special Publication. Urea solutions that are not made of urea and water of the appropriate quality and cleanliness may damage the SCR system. Poor or questionable quality DEF can lead to additional repair and maintenance costs to the customer. Cat warranties do not cover failures caused by or related to use of out of specification urea solutions in Tier 4 Stage IIIB MLIT Step 4 products equipped with SCR systems.

Materials compatibility

DEF is corrosive. Due to the corrosion caused, DEF must be stored in tanks constructed of approved materials. Recommended storage materials:

Stainless Steels:

- 304 (S30400)
- 304L (S30403)
- 316 (S31600)
- 316L (S31603)

Alloys and metals:

- Chromium Nickel (CrNi)
- Chromium Nickel Molybdenum (CrNiMo)
- Titanium

Non-metallic materials:

- Polyethylene
- Polypropylene
- Polyisobutylene
- Teflon (PFA)
- Polyfluoroethylene (PFE)
- Polyvinylidene fluoride (PVDF)
- Polytetrafluoroethylene (PTFE)

Materials NOT compatible with DEF solutions include aluminum, copper, copper alloys, magnesium, zinc, nickel coatings, silver, and carbon steel and solders containing any of the above. Unexpected reactions may occur if DEF solutions come in contact with any non-compatible material or unknown materials.

Bulk storage

Follow all local regulations covering bulk storage tanks. Follow proper tank construction guidelines. Tank volume typically should be 110% of planned capacity. Appropriately vent indoor tanks. Plan for control of overflow of the tank. Heat tanks that dispense DEF in cold climates.

Bulk tank breathers should be fitted with filtration to keep airborne debris from entering the tank. Desiccant breathers should not be used because water will be absorbed, which potentially can alter DEF concentration.

Handling

Follow all local regulations covering transport and handling. DEF transport temperature is recommended to be −5° C (23° F) to 25° C (77° F). All transfer equipment and intermediate containers should be used exclusively for DEF. Containers should not be reused for any other fluids. Ensure that transfer equipment is made from DEF-compatible materials. Recommended material for hoses and other non-metallic transfer equipment include:

- Nitrile Rubber (NBR)
- Fluoroelastomer (FKM)
- Ethylene Propylene Diene Monomer(EPDM)

The condition of hoses and other nonmetallic items that are used with DEF should be monitored for signs of degradation. DEF leaks are easily recognizable by white urea crystals that accumulate at the site of the leak. Solid urea can be corrosive to galvanized or unalloyed steel, aluminum, copper, and brass. Leaks should be repaired immediately to avoid damage to surrounding hardware.

Cleanliness

Contaminants can degrade the quality and life of DEF. The recommended DEF cleanliness target as dispensed into the machine or engine fill tank is ISO 18/16/13. Refer to the Contamination Control Chapter in this Special Publication.

PROOF

Filtering DEF is recommended when dispensed into the DEF tank to achieve the recommended cleanliness targets. DEF filters should be compatible with DEF and should be used exclusively with DEF. Check with the filter supplier to confirm compatibility with DEF before using. Mesh type filters using compatible metals, such as stainless steel, can be used. Paper (cellulose) media and some synthetic filter media are not recommended because of degradation during use. The filter size must be appropriate for the DEF pump design, flow rate, expected pressures and required filter life. Refer to Table 78 below for filtration requirements for Cat engines.

For large machine engines including 3500 series and C175 engine models, filtration of DEF as it is dispensed from bulk tank is recommended to support the durability designed into the SCR system. Filters should be located between the bulk DEF tank and the ground level DEF receiver nozzle.

Note that Cat Diesel Exhaust Fluid Filter group 491-6779, 40 microns efficiency and Beta value 75 can be used to filter DEF when applicable. Refer to your local Cat dealer for availability.

Care should be taken when dispensing DEF. Spills should be cleaned immediately. Machine or engine surfaces should be wiped clean and rinsed with water. Caution should be used when dispensing DEF near an engine that has recently been running.

Note: Spilling DEF onto hot components may cause the release of ammonia vapors. Do not breathe ammonia vapors. Do not clean up any spills with bleach.

Table 78

DEF Filtration Requirements For Bulk Tanks in Specific Applications		
DEF System	**Filter Requirements (1)**	**Filter location**
All Cat engines except those given below in this Table.	Recommended: ≤ 5 microns (µm), Beta value ≥ 1000 Required: ≤ 10 microns (µm), Beta value ≥ 1000	Inlet to DEF Pump Electronic Tank Unit (PETU)
Airless 3500 series and C175 engine models Machine engines (Fast or conventional fill)		
Airless Marine applications		
Air Assisted, Marine applications	≤ 40 microns (µm), Beta value ≥ 1000	Inlet to DEF Dosing Cabinet
Air Assisted, Locomotive and Electric Power Generation applications		

(1) Refer to Caterpillar publication PELJ0133, “Filters, Use of Micron Ratings in the Filter Industry” for clarification of Beta values and filter efficiency

PROOF

Stability

DEF fluid is stable when stored and handled properly. The quality of DEF rapidly degrades when stored at high temperatures. The ideal storage temperature for DEF is between −9° C (15.8° F) and 25° C (77° F). DEF that is stored above 35° C (95° F) for longer than 1 month must be tested before use. Testing should evaluate Urea Percentage, Alkalinity as NH3 and Biuret content.

The length of storage of DEF is listed in the following table:

Table 79

Storage Temperature	Expected DEF Life
Below 25° C (77° F)	18 months
25° C (77° F) to 30° C (86° F)	12 months
30° C (86° F) to 35° C (95° F)	6 months
Above 35° C (95° F)	test quality before use

Refer to “ISO 22241” document series for more information about DEF quality control.

Note: Dispose of all fluids according to applicable regulations and mandates.

General Characteristics of DEF

For detailed information on the requirements and characteristics of DEF, refer “ISO 22241”. For a quick reference, typical characteristics of DEF are given in Table 80 .

Table 80

Characteristics for Urea Solutions		
Property	Unit	DEF 32.5 percent
Urea content		32.5 percent (1)
Alkalinity as NH_3	Percent	0.2
Density at 20° C (68° F)	g/L	1.087 - 1.093 (2)
Refractive Index at 25° C (77° F)		1.381 - 1.384 (3)
Biuret	Percent	0.3 max
Aldehydes	mg/kg	5 max
Insoluble Matter	mg/kg	20 max
Aluminum	mg/kg	0.5 max
Calcium	mg/kg	0.5 max
Chromium	mg/kg	0.2 max
Copper	mg/kg	0.2 max
Iron	mg/kg	0.5 max

(continued)

PROOF

(Table 80, contd)

Characteristics for Urea Solutions		
Magnesium	mg/kg	0.5 max
Nickel	mg/kg	0.2 max
Phosphate (PO_4)	mg/kg	0.5 max
Potassium	mg/kg	0.5 max
Sodium	mg/kg	0.5 max
Zinc	mg/kg	0.2 max

(1) Acceptable range is 31.8 - 33.2 percent
(2) Target value is 1.090 g/L
(3) Target value is 1.382

PROOF

Contamination Control

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Contamination Control

SMCS Code: 1280; 1348; 3080; 5095; 7581

Defining Contamination

Contamination is defined as the presence of unwanted foreign substances in fluid systems or fluid wetted parts. Contamination alters the properties of fluids, causes damage of fluid systems, and prevents systems and components from attaining the desired reliability and durability. Contamination is the primary cause of fluid system failures.

Contaminants include a wide variety of unwanted substances including but not limited to the following:

- Foreign and abrasive substances such as wear particles, fibers, dirt, and dust
- Chemical substances such as products of combustion that are suspended in the fluids
- Cross contamination of water, coolant, oil, and fuel
- Biological micro-organisms such as algae or fungi
- Physical/chemical contaminants such as products of oxidation and heat

Some contaminants are generated within the fluid system due to the normal operation of the system. Contaminants may be drawn into the system from the outside environment or contaminated fill fluids or improper maintenance and repair practices.

Particle contaminants are visible to the naked eye if the particles are approximately 40 µm (microns) and larger while smaller particles are not visible. Particle contaminants can cause damage even if the particles are not visible to the naked eye. The critical particle size for wear particles in a modern diesel engine fuel system is 4 µm.

Contaminants of all types can be controlled by following contamination control practices and using appropriate filtration. Refer to your Operation and Maintenance Manual and to your local Cat dealer for recommendations.

Controlling contamination is especially important for current machine systems. Current machine systems such as hydraulic systems and fuel injection systems are designed with close tolerances and operate at high pressures for enhanced performance. These design improvements emphasize the importance of higher performing fluids, enhanced fluid filtration, and greatly improved fluid cleanliness levels.

Measuring Cleanliness

Fluid cleanliness can be measured by taking fluid samples from various machine compartments. Your Cat dealer can analyze the samples. Particle contaminants are typically measured by particle counters. Chemical contaminants can be measured by specific analysis techniques such as oxidation, water, or soot tests. Some chemical contaminants, such as water in fuel, can interfere with the particle counters and can be counted as particles. Refer to your Cat S·O·S lab or to your Cat dealer for more information.

The number of particles in fluids is expressed in “ISO (International Organization for Standardization)” ratings. “ISO 4406” Standard classifies fluid cleanliness by the number and size of particles in 1 milliliter of fluid. “ISO 4406” Standard measures particle size in µm (microns) and reports the resulting count in three code ranges X, Y & Z. The three code range defines the size and distribution of particles in 1 milliliter of fluid:

- The first code range, X represents the number of particles equal to or larger than 4 µm per milliliter of fluid.
- The second code range, Y represents the number of particles equal to or larger than 6 µm per milliliter of fluid.
- The third code range, Z represents the number of particles equal to or larger than 14 µm per milliliter of fluid.

An example of an “ISO 4406” particle count is 18/16/13. Cat “ISO” cleanliness recommendations are expressed as two or three codes, depending on the machine system. The three code range follows “ISO 4406” definitions and is used for liquid fuels such as diesel and gasoline. The two code system, example “ISO -/16/13”, is used for certain lubricant systems. In the two code system, the first number is the number of particles equal to or larger than 4 µm per milliliter of fluid. This number is not required and may be represented by a dash (-). The second number (Y) and the third number (Z) follow “ISO 4406” definitions. Cat reports the Y and Z codes for lubricating oils to keep consistency with older data and reports.

An example of the particle size and distribution of the “ISO 4406” codes is given in Table 81 .

PROOF

Table 81

ISO 4406 Code	Number of particles in 1 milliliter of fluid		
	4µm and up	6µm and up	14µm and up
“ISO 18/16/13”	1300 - 2500	320 - 640	40 - 80
“ISO 21/19/17”	10000 - 20000	2500 - 5000	80 - 160

Note: Several factors affect the results of particle counts. The factors include the cleanliness of the equipment used to obtain the sample, sample techniques, the cleanliness, and type of sample container, particle counter accuracy (calibration, maintenance, and process), and the environment where the sample is procured. Samples should be taken at representative locations in the fluid circulation system or the fluid distribution system when possible. The sample should be protected adequately from contamination during transport to the lab for analysis.

In addition, particle counters may count water droplets and air bubbles as particulate contamination.

Note: American Society for Testing and Measurement has developed “ASTM D7619” “Standard Test Method for Sizing and Counting Particles in Light and Middle Distillate Fuels, by Automatic Particle Counter”. This test procedure was developed in 2010 to count and measure the size of dispersed dirt particles, water droplets, and other particles in 1-D and 2-D diesel fuels when the specified particle counter is used. “ASTM D7619” is also applicable to biodiesel fuels.

Cleanliness Standards for Machine Systems

Cat recommends that machine systems be maintained at the factory defined fluid cleanliness targets.

Cat has established minimum fluid cleanliness targets for fuels and fill oils and for machine roll-off. Fluids filled into the machine or engine fill tanks are recommended to be at the target levels provided in Table 82 or cleaner. Cleanliness targets for applicable machine component systems are referred to as “Roll-off”. Roll-off is defined as the cleanliness specification of the fluid that is to be obtained before the machine returns to work after maintenance and or system invasion repair. When system fill fluids and Roll-off are maintained at or cleaner than the “ISO” cleanliness targets, contamination-related effects will be reduced.

Table 82

Cat Recommended Fluid Cleanliness Targets(1)		
Cat Recommended Cleanliness Targets for Fluids Dispensed into Machine or Engine Fill tanks	Fill oils(2)(3)	ISO -/16/13
	Dispensed fuels	ISO 18/16/13
	Dispensed DEF	ISO 18/16/13
Cat Recommended Machine Roll-off Cleanliness Targets	Hydraulic systems (Implement & Steering)	ISO -/18/15
	Electronic Transmissions	ISO -/18/15
	Mechanical Transmissions	ISO -/21/17
	Differentials, Wheels, and Axles(4)	ISO -/18/15

(1) The fluids should meet or exceed the cleanliness requirements of the listed ISO levels.
(2) For engine oils optical particle counters may not be effective. Instead filter the oil prior to dispensing into the engine tank, use engine oil filters of 12 micron absolute efficiency and ensure that the oil temperature is 20° C (68° F) or higher. Refer to the details given in this article.
(3) For transmission, gear, differential and axle oils, additives and the viscosity of the oil may interfere with particle counting. An alternative is to use adequate filtration to ensure clean oils prior to filling in the machine compartment.
(4) This cleanliness standard applies only to the Series 700 family of rigid frame trucks, 777 size and larger.

The “fill” fluids cleanliness target is not a fluid “delivery” target. The level of cleanliness for delivered fluids is not specified by Cat. Customers can work with the distributors or carriers to determine the cleanliness level of delivered fluids. However, a more effective and economic means to achieve the fill cleanliness targets is to filter the fluids prior to filling into machine tanks as compared with specifying delivery fluid cleanliness level. Follow the guidelines provided in this Contamination Control article.

Although older technology machines may not be able to maintain the recommended cleanliness targets of advanced models, the same contamination control intervention measures such as filtration and subsequent service procedures should be used on all Cat products.

The viscosity and additives of powertrain oils including transmission, gear, differential, and axle oils can interfere with particle counting. An alternative option is to filter the oils using adequate filtration to ensure clean oils prior to filling in the machine compartments.

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Note: Particle counting of new multi-viscosity engine oils may not be effective to assess their cleanliness level. Optical particle counters cannot distinguish between particulate contaminants and additives. Instead, filter the new engine oils as described below. Additionally do not use optical particle count for the evaluation of used engine oils because soot levels render oil too dark for optical particle counters. Soot levels in used engine oils should be evaluated by using S·O·S Services Oil Analysis.

When filtering engine oil before dispensing into the engine tank or when engine oil kidney looping filtration is done, follow these recommendations:

- Use engine oil filters of 12 microns absolute efficiency. Cat Ultra High Efficiency Lube filter is recommended. Consult your Cat dealer for the most current part number.
- Ensure that the temperature of engine oil is 20° C (68° F) or higher.

Consult your Cat dealer for information and solutions to your oil and fuel analysis needs.

General Contamination Control Recommendations or Practices

Maintaining a low contamination level can reduce down time and can control the maintenance cost of the machine. The productive life as well as the reliability of components and fluid systems is often increased as a result of proper contamination control practices.

The following are general guidelines for controlling contaminants.

- Refer to the Recommendations for Fuel Systems in this chapter for recommended fuel cleanliness levels and guidelines.
- Refer to the machine Operation and Maintenance Manual for the required maintenance for all machine compartments.
- When you add oil to a machine, use adequate filtration in order to clean the oil to meet the targets provided in Table 82 .
- Perform scheduled S·O·S Services Oil Analysis for contamination in order to maintain the recommended ISO cleanliness level of fill and machine fluids. Refer to the S·O·S Oil Analysis section in this Special Publication. The particle count analysis can be performed by your Cat dealer. Particle count can be conducted during the scheduled S·O·S Services Oil Analysis for the compartment. Extra oil samples are not required for the particle count sampling.
- Use only coolants that are recommended by Cat for your machine. Follow the recommended maintenance procedure for the cooling system in the Operation and Maintenance Manual for your machine.
- Maintain the engine air filters and air intake system to avoid unwanted contaminant ingression.
- Follow contamination control practices for the shop area, component/machine disassembly areas, parts, shop tools, test setups, test areas, storage areas and waste collection areas, Keep components clean during inspection, assembly, testing, and filling machines with clean fluids. Good practices will enhance component life and reduce downtime associated with contaminants. Your Cat dealer can provide details on proper contamination processes and practices.
- Follow contamination control practices for the workplace and for the worksite. Maintaining clean oil fill fluids saves time and effort and ensures that fill fluids are at the proper cleanliness levels.
- Use properly designed and maintained bulk storage fluids tanks.
- Protect the fluids storage tanks from dirt and water entry by using 4 µm or less absolute efficiency breathers with the ability to remove water.
- Keep the areas around the tanks filler necks clean of debris and water.
- Drain the storage tanks from water and sediments frequently. The draining schedule depends on use of proper inlet and outlet filters, the use of 4 µm breathers with the ability to remove water, and following recommended contamination control practices. Based on the contamination control program followed, and/or on the fuel supplier recommendations, the storage tank draining schedule may be as frequent as daily until no water is present, and then can be extended to longer periods.
- Install and maintain a properly designed and grounded filtration system. Filtration should include at the entry and at the dispensing point. Continuous bulk filtration may be required to ensure that dispensed oils meet the cleanliness target.
- Cover, protect, and ensure cleanliness of all connection hoses, fittings, and dispensing nozzles.

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Note: Bulk fuel filtration units are available through your Cat dealer. Proper maintenance practices of the bulk filtration systems are available through your Cat dealer.

General Recommendations and Contamination Control Guidelines for Fuels

Follow all applicable industry standards and all applicable governmental, environmental, and safety guidelines, practices, regulations, and mandates.

Note: These general recommendations and guidelines concerning maintenance and care of fuel and fuel storage systems are not intended to be all inclusive. Discuss proper fuel safety and health, handling, and maintenance practices with your fuel supplier. Use of these general recommendations and guidelines does not lessen the engine owners and/or fuel supplier responsibility to follow all industry standard practices for fuel storage and for fuel handling.

Note: Where recommendations for draining water and/or sediment and/or debris are stated, dispose of this waste according to all applicable regulations and mandates.

Note: Caterpillar filters are designed and built to provide optimal performance and protection of the fuel system components.

Clean fuels, as detailed below, are strongly recommended to allow optimal performance and durability of the fuel systems and to reduce power loss, failures, and related down time of engines.

Fuels of “ISO 18/16/13” cleanliness levels or cleaner as dispensed into the engine or machine fuel tank should be used. Reduced power, failures and related downtime can result if clean fuels are not used. Fuels of “ISO 18/16/13” are particularly important for new fuel system designs such as Common Rail injection systems and unit injection systems. These new injection system designs utilize higher fuel pressures and are designed with tight clearances between moving parts to meet required stringent emissions regulations. Peak injection pressures in current fuel injection systems may exceed 30,000 psi. Clearances in these systems are less than 5 µm. As a result, particle contaminants as small as 4 µm can cause scoring and scratching of internal pump and injector surfaces and of injector nozzles.

Water in the fuel causes cavitation, corrosion of fuel system parts, and provides an environment where microbial growth in the fuel can flourish. Other sources of fuel contamination are soaps, gels, or other compounds that may result from undesirable chemical interactions in the fuels. Gels and other insoluble compounds can also form in biodiesel fuel at low temperatures or if biodiesel is stored for extended periods. An indication of microbial contamination, detrimental fuel additives interactions, or cold temperature gel is very rapid filter plugging of bulk fuel filters or machine fuel filters.

To reduce downtime due to contamination, follow these fuel maintenance guidelines in addition to the recommendations given in the “Contamination Control” Chapter in this Special Publication:

- Use high-quality fuels per recommended and required specifications (refer to the “Fuel” chapter in this Special Publication).
- Do not add new engine oil, waste engine oil or any oil product to the fuel unless the engine is designed and certified to burn diesel engine oil (for example Caterpillar ORS designed for large engines). Engine oils may raise the sulfur level of the fuel and may cause fouling of the fuel system and loss of performance. Engine oils in fuels can also reduce the maintenance intervals of aftertreatment devices in Tier 4 machines.
- Use recommended Cat filtration products, including Cat Advanced Efficiency Fuel Filters. Change your fuel filters per recommended service requirements or as needed. Never fill the new secondary and tertiary fuel filter with fuel before installation. Use the fuel priming pump to remove air from the system.
- Follow proper practices of fuel transport and filtration from storage tank to the machine to allow the delivery of clean fuel to machine tank. Keep the fuel storage tank clean of water, debris, and sediment.
- Filter the fuel coming into the bulk storage fuel tank and at every subsequent transfer into and out of any container and prior to adding to the engine fuel tank preferably through filters with a rating of 20 microns (c) absolute or less. The use of wire mesh media (strainer-type filters) is NOT recommended except when filters with standard media (cellulose or synthetic) are downstream of the wire mesh media filters. Wire mesh filters typically have poor filtration efficiency and can corrode with time, allowing the passing of large particles.

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- Caterpillar recommends the use of properly designed and grounded bulk fuel filter / coalescer units which remove both particulate contamination and water in a single pass. These units can clean fuel to “ISO 16/13/ 11” or cleaner and can remove free water to 200 ppm (mg/kg) or less.
- Cat offers heavy duty filter / coalescer units to accommodate fueling rates from 50 to 300 gpm (gallons per minute). Cat custom designs filter / coalescer units specifically for the conditions of fuel at the worksite if needed. Refer to Special Publication, PEHJ0156, “Cat Bulk Fuel Filtration Systems”, and consult your Cat dealer for availability of bulk filtration products.
- Fill machine fuel tanks with fuels of “ISO 18/16/13” cleanliness level or cleaner, in particular for engines with common rail and unit injection systems. When you refuel the machine, filter the fuel through a 4 µm absolute filter (Beta = 75 up to 200 at 4 microns) to reach the recommended cleanliness level. This filtration should be located at the device that dispenses the fuel to the engine or machine fuel tank. In addition, filtration at the dispensing point should have the ability to remove water to ensure that fuel is dispensed at 200 ppm water or less.
- Keep the area around the fuel tank filler neck clean of debris to prevent dirt entry and contamination of the fuel tank.
- Drain your water separators daily per the Operation and Maintenance Manual of your machine.
- Install desiccant type breathers of 4 µm or less absolute efficiency with the ability to remove water on bulk storage tanks.
- Drain your fuel tanks of sediment every 500 hours or 3 months per the Operation and Maintenance Manual of your machine.
- Centrifugal filters may need to be used as a prefilter with fuel that is severely contaminated with gross amounts of water or large particulate contaminants. Centrifugal filters can effectively remove large contaminants but may not be able to remove the very small abrasive particles required to achieve the recommended “ISO” cleanliness level. Bulk filter / coalescers are necessary as a final filter to achieve the recommended cleanliness level.
- Cover, protect, and ensure cleanliness of all connection hoses, fittings, and dispensing nozzles.
- Test for microbial contamination regularly and take proper corrective action if contamination is present. Properly dispose of cleanup waste according to all applicable local regulations and mandates.
- Every 3 months, or sooner if problems are suspected, analyze the fuel for acid number, density, particle content, water and microbial growth (tank bottom fuel sample). More tests can also be run per the “Caterpillar Specification for Distillate Diesel Fuel for Off-Highway Diesel Engines” table in this Special Publication. Refer to Caterpillar S·O·S Services Fuel Analysis Section of this Chapter. Take corrective action if necessary. Corrective actions may include, but are not limited to, treating the fuel, cleaning of the fuel storage tank/system, and replacing the problematic fuel with fresh fuel.
- When fuels are stored for extended periods, follow all the fuel supplier and tank maintenance procedures. Circulate the fuel regularly through a filter to remove sediments. Test the fuel regularly for acid number, density, particle content, water and microbial growth. Observe trends of these properties to ensure no detrimental changes. The fuel is not recommended for use when its properties change negatively. For fuels containing biodiesel, the storage duration may be significantly reduced. Observe all the guidelines given in this section and in "Guidelines and potential impacts associated with the use of biodiesel and biodiesel blends" Table in the "Biodiesel" section of this Special publication.

NOTICE

In order to meet expected fuel system component life, 4 micron(c) absolute or less secondary fuel filtration is required for all Cat diesel engines that are equipped with common-rail fuel systems. Also, 4 micron(c) absolute or less secondary fuel filtration is required for all Cat diesel engines that are equipped with unit injected fuel systems. For all other Cat diesel engines (mostly older engines with pump, line and nozzle type fuel systems), the use of 4 micron(c) absolute or less secondary fuel filtration is strongly recommended. Note that all current Cat diesel engines are factory equipped with Cat Advanced Efficiency **and/or Ultra High Efficiency** filters 4 micron (c) absolute fuel filters.

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